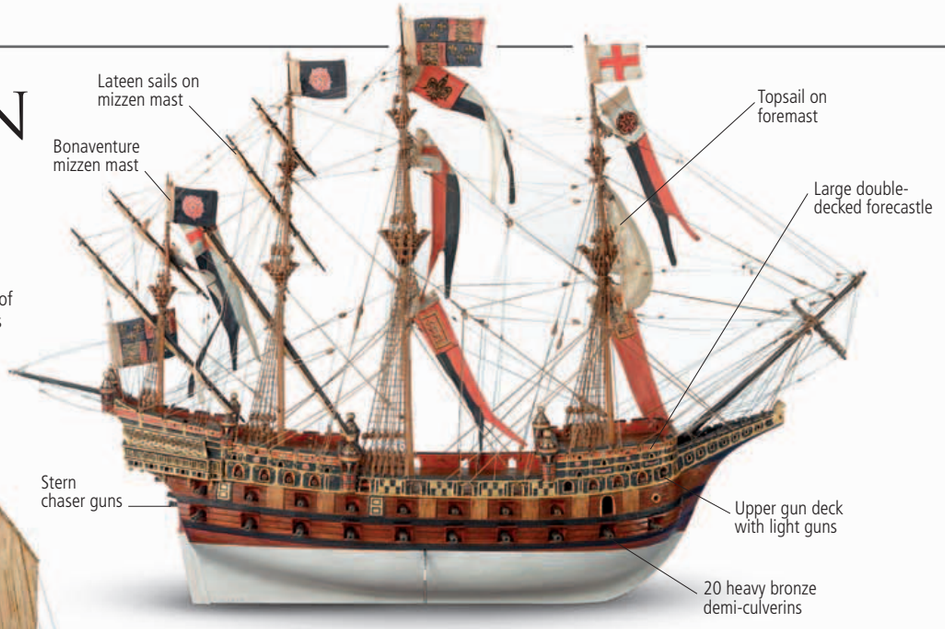
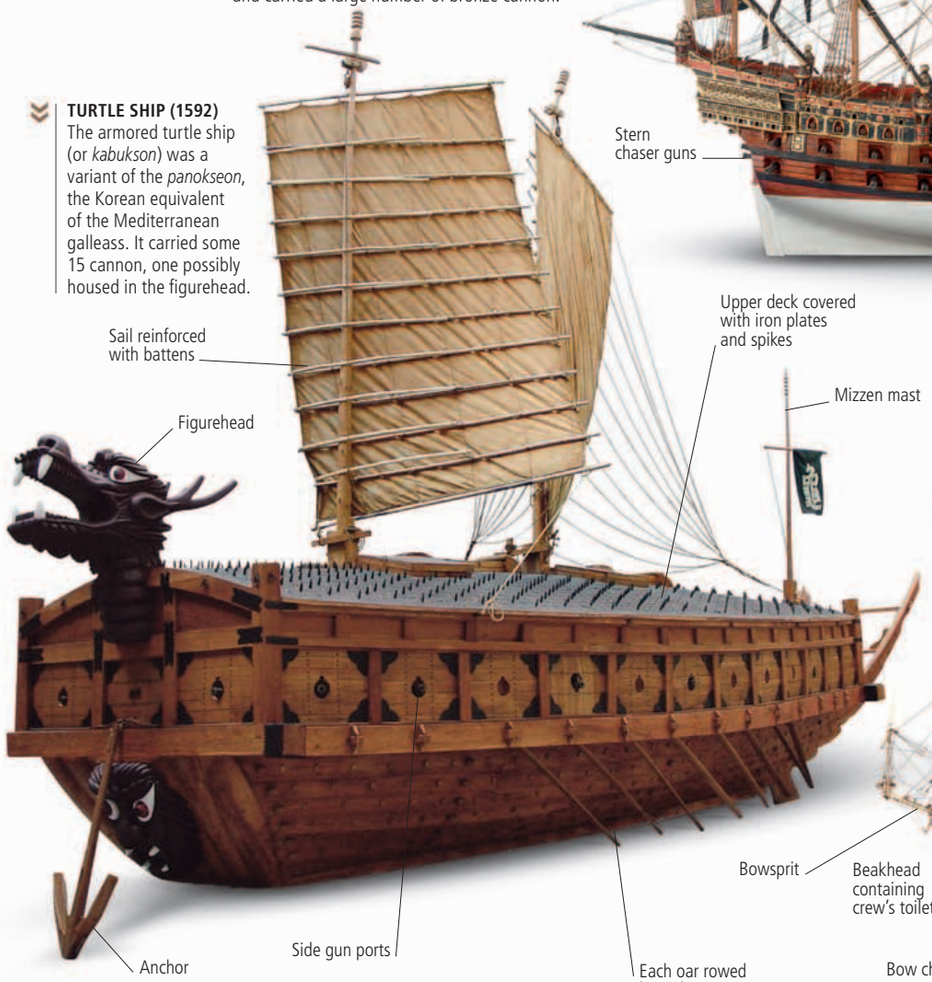
EARLY MODERN WARSHIPS

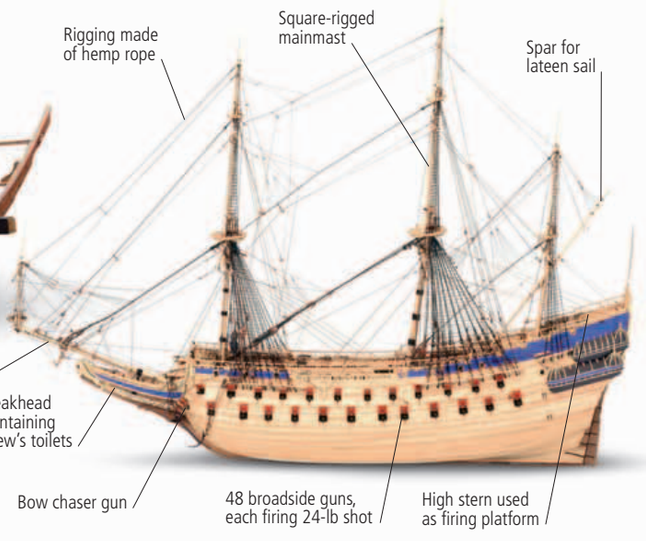
GALLEON (1560) In 16th-century Europe the carrack evolved into the sleeker galleon, which established itself as the most effective warship of its day. Galleons usually had three or four masts and carried a large number of bronze cannon.



TURTLE SHIP (1592) The armored turtle ship (or *kabukson*) was a variant of the *panokseon*, the Korean equivalent of the Mediterranean galleass. It carried some 15 cannon, one possibly housed in the figurehead.



GREAT SHIP (1628) In the 17th century European monarchs had large elaborately decorated ships built for prestige rather than military efficiency. The 64-gun *Vasa* was constructed on the orders of the ambitious Swedish king Gustavus Adolphus. It sank on its maiden voyage.



THE DEVELOPMENT of ocean-going sailing ships in the 16th century allowed European sailors to make voyages around the world. They encountered countries with independent and highly sophisticated naval traditions. For example, in a war against Japan in the 1590s, the Korean fleet included ironclad "turtle ships," the likes of which had never been seen in Europe. China had built sailing ships larger than any that Europeans possessed. Yet the momentum of European progress in shipbuilding and arms manufacture was unstoppable.

GALLEONS AND GALLEASSES

In the second half of the 16th century the large carrack or "great ship" was surpassed as a fighting warship by the galleon. Galleons had all the advantages of carracks—they were substantial-size ships with gun ports that could carry a heavy weight of cannon—but their designers reduced the height of the castles and lengthened and streamlined the dimensions of the ships for better handling. The sides of the ships were sloped inward from the lower gun deck to the weather deck, enhancing stability and making them more difficult to board.

While galleons dominated warfare in the Atlantic, in the Mediterranean oared galleys remained in use in battle through the 16th and 17th centuries. The Spanish and Venetians turned galleys into floating gun platforms. Venice invented the galleass, a large warship powered by sails and oars, which could mount a truly impressive quantity of cannon. But although galleasses were used outside the Mediterranean—they formed part of the Spanish Armada in 1588—the future lay with ships that relied entirely on sails.

SHIPS OF THE LINE

In the course of the 17th century the galleon evolved into the ship of the line. Gunnery became the primary means of engaging enemy vessels and the traditional action of boarding only secondary. Guns were arranged almost exclusively along the sides of the ship, to deliver thunderous broadsides. Ships fought in line of battle, and the ship of the line was a warship powerful enough to hold its own in such a gun duel. The forerunner to such ships was England's *Sovereign of the Seas*, commissioned by King Charles I of England in 1634. Smaller frigates and sloops played a supporting role.